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Anthropometry, bioimpedance and densitometry: Comparative methods for lean mass body analysis in elderly outpatients from a tertiary hospital

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ABSTRACT

Objectives: To evaluate lean mass index (LMI) measured by bioimpedance (BIA) and anthropometry compared to densitometry (DXA) in elderly outpatients from a tertiary care hospital.

Methods: Participants were over 60-year-old men, presenting no dementia or disability, from a tertiary geriatric ambulatory. LMI obtained by BIA, anthropometry and DXA were submitted to Baumgartner, Janssen and Delmonico calculations respectively. Sarcopenia was calculated as LMI by DXA and handgrip strength. Data were analyzed by T student's test, ANOVA for repeated measures and pos hoc Bonferroni test, Pearson's correlation test, regression equation and Bland Altman analysis, ROC curve and contingency table 2 × 2 for sensitivity, specificity and predictive values.

Results: A total of 92 participants completed the study. Most of them were married, aged 72.9 ± 6.6, lived a sedentary lifestyle, presented multiple morbidities, and in use of polypharmacy. Appendicular lean mass was lower in sarcopenic participants when compared to that in nonsarcopenic ones (20.2 kg/m^2 and 23 kg/m^2 respectively, $p < 0.0001$). BIA sensitivity, specificity and correlation to DXA were 37%, 98% and $r = 0.81$ ($p < 0.001$), and for anthropometry 67%, 92% and $r = 0.77$ ($p < 0.0001$) respectively. Bland Altman's analysis showed congruence between methods and DXA (anthropometry: bias = −0,05 ± 0,66, limits of agreement (LoA) = −1.37 and 1.26; BIA: bias = 2,2; LoA = 0,7 and 3,7).

Conclusion: Aging and multiple chronic and degenerative morbidities affect LM in vulnerable elderly patients. Both anthropometry and BIA, are accurate to measure LMI independently in this population but Anthropometry presented better agreement to DXA than Bioimpedance and has the advantage of lower price, easier application and cheaper equipment to be applied.

1. Introduction

The global population is progressively aging in an accelerated pattern, making the elderly's quality of life a main concern for political, social and economic strategies towards individual and collective health demands (Zanon et al., 2013).

The aging process demands changes in body composition, such as the gradual metabolic and hormone declination, continuous muscle mass substitution for fat tissue, osteoporosis, joint pain, less thirsty that leads to dehydration (Zanon et al., 2013). Several authors studied the imbalance of body fluids in the elderly based on concepts of hyper and dehydration or hypovolemia. It is stated that dehydration is the most common water disorder, leading to increased morbidity and mortality, as well as prolonged hospital stay in elderly patients. In older individuals, thirst sentation, as a physiological response to water deprivation and infusion of hypertonic saline solution, is decreased, osmotic point for feeling thirsty and fluid replacement is slower in the elderly. They present increased relative amount of body fat, reduced relative body water volume, higher proneness to disorders. In addition to impaired sodium excretion, decreased secretion of renin, angiotensin and aldosterone as they age. as well extremely hight release of the antidiuretic hormone. There are clinical circumstances that increase susceptibility to dehydration (Ritz and for the Source S, 2001; Rösler et al., 2010).

In addition, multiple degenerative chronic diseases are frequently associated to this population. These morbidities induce chronic continuous underlying inflammation and may potentiate vulnerability (Zanon et al., 2013). All these facts point out the pivotal role of keeping and consistently evaluating the level of autonomy, independence and sarcopenia among this population (Chatterji et al., 2015; Beard et al.,

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2016; Cruz-Jentoft et al., 2010; Landi et al., 2013).

Sarcopenia is defined as low lean mass and low muscle strength. It is one of the most frequent, costly and incapacitating diseases during the geriatric period. The physiopathological mechanisms are multifactorial and like aging (Cruz-Jentoft et al., 2010). Sarcopenia is associated with the negative health outcomes in the geriatric population, such as high prevalence of multiple diseases, early death and severe disabilities and dependence for performing daily activities (Morley et al., 2011; Rosenberg, 1997).

Therefore, early accurate diagnose of sarcopenia is mandatory, mainly in the vulnerable elderly. However, analytical techniques used for measurement of LMM (lean muscle mass) in young subjects may not be valid in this population because of different water and tissue in body composition (Ribeiro and Kehayias, 2014; Lustgarten and Fielding, 2011).

Dual x-ray absorptiometry (DXA) measures bone mass and quality, fat and lean mass, as well as their total and segment percentage distribution (Heymsfield et al., 1997; Vilaca et al., 2011). DXA, computed tomography and magnetic resonance are considered gold standards for estimating lean mass. Notable advantages of DXA include accuracy, simplicity, availability, speed of measurement (whole-body scans require less than 20 min), exposure to low levels of radiation (< 1 mrem), and the ability to perform fast serial-section measurements. LMM (lean muscle mass) measurements by DXA have been validated for use in elderly subjects, when measurements of total body 40 K and nitrogen were used as the reference standard, and provides evidence that body cell and skeletal muscle mass decline at the same rate during aging (Lustgarten and Fielding, 2011). However, DXA is expensive, not easily available in the health public system, achievable just for people below 150 kg and requires trained professionals to be performed (Vilaca et al., 2011; Guglielmi et al., 2016).

An alternative method is electric bioimpedance (BIA), a non-invasive portable equipment, easily handled and reproducible. It directly quantifies the body composition fraction between fat and lean mass (Eickemberg et al., 2011). BIA measurements of body composition are based on the concept that tissues rich in water and electrolytes are less resistant to the passage of an electrical current than lipid-rich adipose tissue. So, hydration status is a key point in BIA estimation of LMM. Hydration in the elderly is critical. So LMI by BIA has shown conflicting results during aging and in other specific populations (Lustgarten and Fielding, 2011).

Anthropometric measurements could also be very practical for the clinical assessment of nutritional status in older people. Anthropometry offers the single most portable, universally applicable, inexpensive, and noninvasive technique for assessing the size, proportions and composition of the human body. Furthermore, anthropometric calculation may include muscle strength, thus allowing the structural and functional diagnosis of sarcopenia (Landi et al., 2014).

However, data on anthropometry has limits because it depends on the ethnicity (Lustgarten and Fielding, 2011) and direct man made muscle and fat body measurements. These measurements may be more difficult in the elderly population due to reduction of muscle and fat in the limbs, loss of elasticity leading to skin folds making it more difficult to separate muscle tissue from adipose tissue, as well as in the measurement of circumferences, especially in the adjustment of the tape measure and in the identification of the correct place to perform the measurement. Therefore, its use is recommended as long as the limitations mentioned above are taken into account (Baumgartner et al., 1991).

Similar to most developing countries, Brazil has a large miscegenated population (210 millions people) and great social gap (human development index- HDI 0,69) (Sociais ICdPeI, 2018). In addition, as in the whole world, Brazilian population is getting old and chronic degenerative diseases are increasing. Estimations of sarcopenia in Brazilian population, either sick or healthy, is rare studied body composition in Brazilian healthy adults aged 18–94 years old and stratified values for age and sex (Barbosa-Silva et al., 2005). Some authors have studied LMI in specific Brazilian populations either healthy (Fregonasse et al., 2003; Fonseca et al., 2018; Ferro et al., 2018). However, there is no study on LMI in the Brazilian elderly vulnerable population with multiple diseases.

Therefore, a study focusing the accuracy of accessible methods to measure LMI in a vulnerable miscegenated old population is mandatory. So, we aim to study the accuracy of 2 methods anthropometry and bioimpedance compared reference standard DXA in an old fragile population from the largest tertiary and quaternary care hospital in South America, in São Paulo, Brazil.

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2. Methods

It is a study of accuracy in outpatients from a tertiary teaching public hospital conducted in 2014 and 2015. This study was approved by the Ethic Committee of Research Project analysis – CAPPesq from Hospital das Clinicas da Faculdade de Medicina da Universidade de São Paulo (HCFMUSP), number CAAE: 27387314.2.0000.0068. All the patients who were included in the study signed the consent form.

The inclusion criteria included over 60-year-old male outpatients attended by a trained professional (geriatric physician) at the Geriatric Division of HCFMUSP. The exclusion criteria included: refusal to be submitted to the study's tests, reduced mobility or bedridden condition, use of cardiac pacemaker, presence of dementia, severe or decompensated diseases, such as cardiac failure functional class III and IV, exacerbated COPD, liver diseases classified as Child B/C, dialytic renal failure, cancer or depression with functional limitation.

The HCFMUSP is a teaching public tertiary and quaternary care hospital, the largest in Latin America. The HCFMUSP Geriatric Division cares for multiple morbidities patients (> 2 diseases), usually presenting multiple organ failures, making use of polypharmacy (> 5 kinds of medications), and coming from across the country. The prevalent diseases are non-infectious chronic degenerative diseases, such as obesity, diabetes, arterial hypertension, thyroid diseases, osteoarthritis, COPD, peripheral neuropathy, dyslipidemia, dementia and prostatic disease.

Social demographic data, morbidities and medications in use were obtained from medical records.

The International Physical Activity Questionnaire (IPAQ), validated in Brazil, was used to evaluate sedentary lifestyle (Sebastião et al., 2012).

The measured data were handgrip strength, anthropometric data, such as weight, height, waist and hip circumference, tricepital skinfold, calf circumference resistance, reactance and appendicular mass, lean mass index (LMI) by bioimpedance Bodystat, and the Hologic Discovery DXA. All these data, except for that of DXA, were collected by the same trained professional at the same time. All DXA data were measured by a specialist. Anthropometric and DXA measurements were made within a ten-day period.

Handgrip strength was measured in the dominant limb. Technically, the participant should be seated, have the shoulders abducted in neutral rotation, the forearm flexed at 90° in neutral position with the wrist extended at 0°, ulnar deviation at 0° to 15°, and the feet aligned on the floor (Norgan, 1988). The average of 3 consecutive measurements by the DETECTO DHS Series – Medical Handgrip Dynamometer was used as each handgrip data (Hillman et al., 2005). Anthropometric data were measured as described by Norgan *et al* (Norgan, 1988).

For Bioimpedance data, the patients were lying down in horizontal dorsal decubitus position with electrodes connected to the right hand and foot. The proximal electrode was connected to the styloid process and the distal electrode was connected to the median finger. The distal electrode was fixed between the lateral and medial malleoli on the inferior limb. The patients were fasted for at least 4 h, did not drink or eat caffeine nor did they do intensive exercises within the last 24-h

period (Eickemberg et al., 2011).

LMI was calculated as suggested by Baumgartner *et al* (Baumgartner et al., 1998) for anthropometry and handgrip, Janssen *et al* (Janssen et al., 2000) for BIA, and Delmonico *et al* (Delmonico et al., 2007) for DXA as follows:

- Baumgartner *et al* (Baumgartner et al., 1998): Appendicular muscle mass (kg) = (0.2487 × weight) + (0.04873 × height) - (0.1584 × hip circumference) + (0.0732 × handgrip strength) + (2.5843 × gender) + 5.8828 e Body fat (%): 0.2034 (waist circumference) + 0.2288 (hip circumference) + 3.6827 (log tricepital skinfold) – 10.9814 (gender) - 14.3341;
- Janssen *et al* (Janssen et al., 2000): muscle mass (kg) = [(hight^2/ Resistance ×,41) + (gender × 3.825) + age × 0.071)] + 5.102 and muscle mass index (MMI) kg/m^2 = muscle mass/ height^2 The low MMI was set at ≤8.87 kg/m^2;
- Delmonico *et al* (Delmonico et al., 2007) appendicular muscle mass (MMa) (kg): −22.59 + 24.21 × height (m) + 0.21 × total fat (kg) and MMa index (kg/m^2) = MMa/height^2. The low MMI was set at < 7.25 kg/m^2.

Lean mass index value calculated by DXA and handgrip strength was measured simultaneously and was used to stratify the participants into sarcopenic and nonsarcopenic groups. The limit for a nonsarcopenic person was set at < 7.25 Kg/m^2 (DXA) and handgrip strength at < 30 kgf.

2.1. Statistical analysis

Data were submitted to statistical analysis by SPSS version 20 and Prisma. Continuous data were expressed as mean and standard deviation and submitted to Levene test (homogeneity), Kolgomorov-Smirnov test (normality). Categorical data were stratified as dicotomic (0 = normal; 1 = altered, except for sex which was stratified as 0 = female and 1 = male). Sarcopenic patients were compared to non sarcopenic patients by Student's *t*-test. The observed differences among FFMM (fat free muscle mass) measuring methods were tested for significance by ANOVA for repeated measures and *post hoc* Bonferroni test as these continuous data are related to the same population. Sphericity to fulfill the assumption of ANOVA repeated measures was checked by Malchlys's test. A ROC curve was built to show the sensibility and specificity of the studied methods. A contingency table 2 × 2 was used for quantifying sensitivity, specificity, and predictive values. The association among methods was checked by Pearson's correlation and the levels of agreement were assessed according to the method of Bland Altman. Bland- Altman analysis were made sequentially as suggested by Hiraka et al. (Hirakata and Camey, 2009): descriptive and normality test for variables (by explore data and Kolgomorov-Smirnow- KS), scatter plot of dependent and independent variables with adjusted line to check variables relationship, mean difference of variables data (bias), mean of variables data and percentage of bias calculation (by transforming data), LoA (limits of agreement) by simple calculation, bias difference from predicted mean difference (zero) by Student's T one sample test, build Bland Altman's plot, bias dependence of the size measurements (by Pearson coefficient), the bias dispersion (bias normality by KS test) and the bias proportional dispersion (by linear regression of bias as dependent value and variables mean as independent variable) all made in SPSS. Significance was set at 5%.

3. Results

All patients that attended by the trained professional at the Geriatric Division of HC-FMUSP during 2014 and 2015 were invited to the study. A total of 208 patients were invited, 125 were recruited and 108 were included in the study. Out of those, 16 were excluded, since they did not complete all the tests.

Table 1
Social demographic data

N = 92		
Schooling	Years of education (mean, SD)	9.6 ± 3.8
Marital status n(%)	Married	75(81.5)
	Single	13(14.1)
	Widower	03(3.2)
	Divorced	03(3.2)
Race n(%)	Caucasian	61(66.3)
	Mulatto	13(14.1)
	Asiatic	11(11.9)
	Black	04(4.3)
Work activity n(%)	Yes	19(20.7)
Lifestyle n(%)	Sedentary	67(72.8)
Diseases n(%)	≥ 2	53(57.6)
BMI (kg/m^2) n(%)	Low	16(17.4)
	Normal	46(50.0)
	Obesity	30(32.6)
Medications n(%)	< 5	78(84.8)
	> 5	14(152)

BMI (kg/m^2): body mass index; N = 92: sample; CC and years of age expressed as mean and standard deviation.

The participants' social demographic characteristics are described in Table 1. The main profile consisted of old men in their seventies, married, with sedentary lifestyle, with central obesity and limb hypotrophy, multiple chronic degenerative diseases and in use of polypharmacy. In Table 2 are described the body composition assessment in older adults.

Concerning mobility and sedentary lifestyle tested by IPAQ, all participants were sedentary and presented multiple morbidities, mainly chronic degenerative diseases.

Predicted values of bioimpedance, show SEE: 0.73 kg and r^2: 059, for anthropometry SEE: 0.74 and r^2:0.57.

Sarcopenia was present in 28.2% (n = 26) patients and they were predominantly older, 76.6 ± 7.6 years old (p = 0.003), with lower handgrip strength, MMa and appendicular muscle mass when compared to non sarcopenic individuals as expected (Hand grip: 24.1 ± 4.7 kgf, p < 0.0001; MMa:17.30 ± 1.96 kfg, p < 0.0001; appendicular muscle mass: 20.2 kg/m^2 and 23 kg/m^2, p < 0.0001, T Student's test) but there was no difference between calf circumference tested by DXA in those patients (36.0 ± 3.3 cm for the right side and 36.27 ± 3.57 cm for the left side, p > 0.84, Student's *t*-test) .

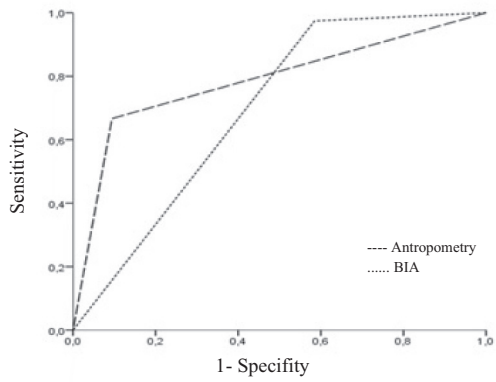
One way ANOVA for repeated measures showed that values presented sphericity (Malchlys's test p = 0,253) and that there was difference among methods (DXA = 7,67 ± 1,14; Anthropometry = 7,61 ± 0,82; BIA = 9,87 ± 1,11, p <0001). Bonferroni test showed that there was no difference between Anthropometry and DXA measurements (p = 0,82) but BIA was different from DXA (p <0,01) and Anthropometry (p < 0,01) presenting

Table 2
Body composition assessment in older adults.

N = 92	Mean	SD
Reactance(ohms)	47.7	± 12.1
Resistance(ohms)	478	± 12
Handgrip strength(kgf)[1]	29.5	± 7.7
Waist circumference (cm)[3]	96.0	± 12.1
Tricepital skinfold (mm)[2]	13.4	± 5.8
Hip circumference(cm)[3]	96.7	± 8.3
Calf circumference (cm)[3]	36.0	± 3,3
Appendicular muscle mass(kg)[4]	21.1	± 3.7

Data are presented as Mean ± SD (standard deviation).
[1] kgf = kilogram force
[2] mm = millimeter
[3] cm = centimeter
[4] kg = kilogram

A



Antropometry: Area under ROC curve:0.84 [CI:0.77 -0.92]
BIA: Area under ROC curve:0.83 [CI: 0.75 -0.91]

B

	Bioimpedance		Anthropometry	
DXA	**%**	**CI**	**%**	**CI**
Sensibility	93	[0.68-0.99]	86	[0.68-0.96]
Specificity	70	[0.58-0.80]	80	[0.69-0.89]
PPV	37	[0.22-0.55]	67	[0.50-0.81]
NPV	98	[0.90-0.99]	92	[0.82-0.97]

Caption: PPV- positive predictive value, NPV – negtive predictive value, CI- confidence interval, DXA- Densitometry

Fig. 1. Roc Curve 1A, sensitivity, specificity and predictive values of lean mass index measured by bioimpedance and anthropometry compared to DXA 1B.

higher absolute values.

Fig. 1 shows ROC curve the sensitivity, specificity and predictive values of both anthropometry and BIA compared to DXA. Lean mass index measured by Anthropometry presented high sensibility and good specificity and the highest probability for positive tests in healthy individuals. Both methods BIA and Anthropometry presented similar results in sensibility, specificity and negative predictive values.

Fig. 2 shows a panel with charts correlation and charts of the agreement by Bland Altman's plots between DXA and anthropometry (2A) and DXA and BIA (2B).

Both BIA and Anthropometry presented linearity and high association to DXA (Fig. 2) which was supported by Pearson's test (Bioimpedance: $r = 0.81, p < 0.0001$; Anthropometry: $r = 0.77$, p < 0.0001).

The linear regression equation of Anthropometry and DXA measurements was expressed as y = 3,09 + 0,59.x. The 95% confidence interval was 0,73–0,87. The coefficient of determination R2 = 0,66 which means that 66% of the variance of DXA is predictable from BIA in this statistical general linear model.

The linear regression equation of measurements done by BIA x DXA was expressed as y = 4,14 ± 0,75.x. The 95% confidence interval ranged from 0,66 to 0,84. The coefficient of determination was $R^2 = 0{,}59$ which means that 59% of the variance of DXA is predictable from BIA in this statistical general linear model.

Bland Altman plots (Fig. 2A and B) and analysis showed the agreement of Anthropometry and Bioimpedance with DXA. All tested variables were parametric and correlated.

Anthropometry agreement tests showed that bias was negative, very close to and not different from the predicted variables mean (bias = −0,05, −0,20%, $p = 0{,}44$, Student's T one sample test). This result shows that there is a proportional variance between Anthropometry and DXA. LoA limits were narrow −1,37 to +1,26, bias was normally distributed (p = 0,097, KS test), and linear regression supported that there was a trend to bias proportionality ($R = 0{,}48$, $R^2 = 0{,}239$, y = 2,62–0,35x, β = −0,48, p = 0,001). Bland Altman's plot (Fig. 2) for Anthropometry reached 96% agreement and only 4 patients did not fall within the confidence interval. The difference among the mean of the data set was negative, thus reinforcing the high agreement among the results and the error probability at 5%. The low variability in low deviation shows the data's homogeneity but still reflects a trend to systematic error below the mean difference and 4 values out of LoA limits.

Bioimpedance agreement tests showed a positive bias, not so close to the variables means but was different from the predicted variables mean (bias = +2,20, 25,4%, $p < 0{,}01$, Student's T one sample test). This result shows that there is a non proportional variance between BIA and DXA values. LoA limits were within 0,7 to 3,7, bias was normally distributed (p = 0,2, KS test), and linear regression showed that there was aleatory with no trend to bias proportionality (R = 0,043, $R^2 = 0{,}02$, y = −0.031 + 0.76x, β = −0,043, p = 0,685). Bioimpedance and DXA Bland Altman's plot (Fig. 2) also reached 96% but error probability was at 25%. The participants, who fell outside the confidence interval, were predominantly obese (IMC > 30 kg/m^2, $n = 5$), similar body composition in percentage of fat.

4. Discussion

Both anthropometry and BIA proved to be reliable and accurate methods for measuring LMI compared to DXA in the elderly with multiple morbidities in use of polypharmacy.

We opted to study a nonrandomized sample of elderly population with multiple diseases due to their vulnerability and because constant subclinical inflammation may contribute to sarcopenia development. The number of participants depended on the trained professional that attended at the Geriatric Division of HCFMUSP. Still, out of more than 200 patients invited, only 128 were included in the study and 92 of them completed all the tests and provided significant data. Bland Altman describes how to choose the sample size to estimate the 95% limits of agreement in Bland Altman's analysis (Bland and Altman, 1986). He says that although sample size calculation should be based on the most precise obtained LoA which is linked to clinical practice, for a sample size of 100, it is possible to estimate bias and LoA with an CI95% around ± 0,34DP (of bias). Further, a similar study from Kim J et al. (Kim et al., 2002) included 93 participants and presented significant results. So we guess that although non probabilistic populations have less power than randomly assigned populations, they are more often than probabilistic populations in clinical research. and may contribute with significant data.

Participants were old, low educated, sedentary, with multiple chronic degenerative cardiovascular and metabolic diseases, central obesity, thin limbs, lower hand grip capacity and attended at a tertiary hospital. Patients presented trend do LMM (lean muscle mass) and almost one third presented sarcopenia. Aging and sarcopenia are correlated to low physical capacity, functional limitation and disabilities, as well as other factors such as: comorbidities, social condition, lifestyle,

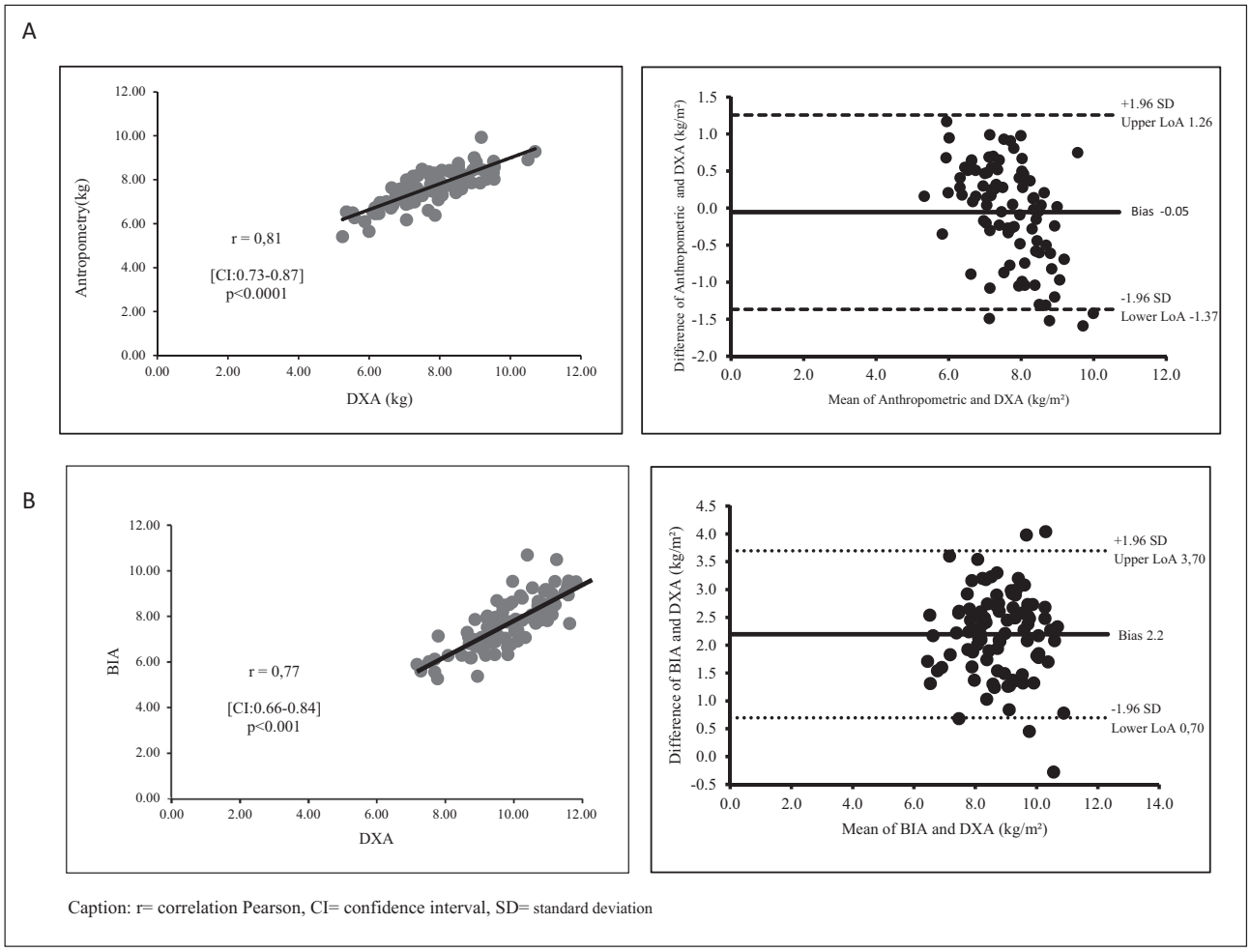


Fig. 2. Relation and agreement between lean mass index measured with DXA and anthropometry data (1A) and BIA data (1B).

falls (Marzetti et al., 2018; Beaudart et al., 2015). In addition, these diseases present vascular atherosclerosis, a progressive and continuous vascular inflammation that may induce energy waste (Dufour et al., 2013). The Framingham study by Dufour *et al* (Dufour et al., 2013) applied DXA to evaluate the LMI (lean muscle index) of 274 old men aged around 79 years old and at risk of cardiovascular outcomes. They found 19% prevalence of sarcopenia (according to Baumgartner *et al*, (Baumgartner et al., 1998) and just 6% with low LMI referred decrease in functionality. Low LMI was associated with a 6 times higher risk of disability (Dufour et al., 2013). Our data showed higher prevalence of sarcopenia than the Framingham study (Dufour et al., 2013). This apparent discrepancy may be due to different definition of sarcopenia as the patients from our study were evaluated by anthropometric and hand grip measurement. Furthermore, the participants of Framingham study differ from those of this study, who belonged to a tertiary and quaternary hospital, presented multiple chronic degenerative diseases, using polypharmacy and are prone to underlying inflammation and energy waste.

This study provided new insights about LMM measurement methods for the vulnerable elderly. There was no difficulty in collecting anthropometric or BIA data independently of age or number of diseases and that there is accuracy in both analyzed methods independently.

This study is based on assumptions of research focusing accuracy by comparing methods: it is a cross section design as a follow up is not needed; it is built on repeated measures of the same population which decreases sample biases; it has a gold standard method which facilitates to state a predicted reliable variable for comparison. The tests were selected sequentially as is used in clinical practice: sensibility, specificity (ROC curve), predicted values (contingency table 2 × 2) and Pearson's coefficient tests (correlation tests) and further agreement tests (as Bland Altman). Correlation tests are easily to perfom and interpret and are very frequently used in clinical practice. They help health professionals to know the probability of being health or sick and for planning which exam is worthwhile to choose.

Anthropometry and BIA were highly positively correlated to DXA by scatter plot, ROC curve and Pearson's coefficient. The positive relation in the correlation line was expected, since all methods measure the same variable, sarcopenia. But there were differences in the comparison and correlation analysis of each method that must be interpreted adequately.

ANOVA repeated measures and pos hoc test showed that BIA measurements were different from and larger than those of DXA and Anthropometry. Anthropometry measurements were not different from those of DXA. This result suggests that BIA overestimates sarcopenia prevalence measurements.

Both BIA and Anthropometry presented high sensibility suggesting to be good screening tests for diagnosing positive (sick) patients. However, Anthropometry presented better positive predictive values and AUC in ROC curve value, thus supporting the hypothesis that they are good screening tests. Regarding specifity and negative predictive values, both methods presented similar results, and the controversy about which method is better to diagnose negative (healthy) individuals remains. So, Anthropometry may be a good recommendation of screening exam for collective or individual approach in diagnosing

sarcopenia in the elderly. This interpretation is not exclusive, but helps the decision making in clinical practice.

Correlation analysis, such as Pearson product-moment coefficient, correlation line, contingency table 2 × 2 and ROC curve are useful for describing the relative strength of a relationship between two variables and their covariance. But regression analysis may provide the basis for predicting this relationship and understanding and predicting clinical outcomes, which may support clinical practice, mainly in situations where resources are scarce.

Both BIA and Anthropometry presented positive and linear regression equations related to those of DXA, positive regression constant different from zero and 95% CI within the limits found in clinical practices, as expected for a clinical experiment. However, Anthropometry presented the lowest regression constant, CI below 95%, while BIA presented a better regression coefficient (slope). These regression constants represent the value of BIA or Anthropometry when DXA is zero. One needs to remember that, in clinical practice, you never get this level of sarcopenia (zero). But these regression constants mean that there is a fixed overestimation between both methods and DXA. Anthropometry presented a lower regression constant, which means lower gap between this method and the gold standard method. This supports the results that Anthropometry measurements were not different from those of DXA, while BIA presented higher fixed absolute values of sarcopenia than those of DXA and Anthropometry. This is important when interpreting results from both methods, since absolute measurements of BIA will probably overestimate sarcopenia levels measured by the standard method.

The regression equation slope (regression coefficient) of both methods are also different from 1.0. If the correlation was equal to one, the change rate of both methods would be the same. So, these results were expected, since both methods are prone to errors. The regression coefficients different from zero mean that the rate of change in units of both methods for each unit of DXA is different. BIA´s regression coefficient is closer to one than that of Anthropometry, thus showing that the proportional rate of change in BIA measurements much similar to that of DXA than those from Anthropometry measurements. This can probably be explained, as both BIA and DXA are directly measured by equipment and so, more prone to random errors. In turn, Anthropometry measurement includes man made measurement and calculations, which are prone to fixed biases. BIA change rate is closer to that of DXA than these of Anthropometry, and this should be taken into account when interpreting measurements of each method. In clinical practice, this may be useful when you have a patient who presents variation on sarcopenia levels the range in BIA's measurement will be more reliable compared to DXA than those of Anthropometry's.

Although the rate of change of BIA is proportionally closer to that of DXA than Anthropometry measurements, the 95% confidence interval of Anthropometry was narrower than that of BIA, and closer to the range of frequently FFMM measurements by DXA. Furthermore, the coefficient of determination (R^2) of Anthropometry is higher than that of BIA. These coefficients of determination translate the percentage of total variance in BIA or Anthropometry that can be explained by DXA. Thus, this parameter is a measure of proportion indicating the accuracy of prediction of both methods against that of the gold standard method. So, the variance of DXA is better predicted by Anthropometry than by BIA.

Correlation tests are reliable and frequently used in clinical practice but have no power to show agreement between methods and may lead to interpretation biases. Agreement tests do not exclude correlation tests but complement it showing the congruence of methods based on the difference (bias) between methods and the error behavior and through the dispersion of errors. Besides, it can infer if the errors are random or systematic, allowing to improve the methodology (Giavarina, 2015).

Bland Altman is a test to check agreement and error behavior. It is based on comparison of methods and difference of variable means tested on the same population by repeated measures, which decreases sample biases (Giavarina, 2015). The results of our study were able to support these facts. Correlation tests were highly positive in both methods, but Bland Altman was able to stratify which of both methods presented more agreement to DXA, the gold standard method.

Bland Altman's analysis on Anthropometry presented high agreement to DXA: small bias close to the observed variables mean difference, Test Student's one test that showed it was no different from the predicted mean difference (zero) and narrow LoA limits so it proved to be a good agreement to the gold standard DXA, and so, to diagnose positively the sarcopenic patients. It also showed trend to non proportional biases, suggesting a predominant error below the mean difference. Looking to Bland Altman's plot this trend in errors is located below the mean difference. This suggests a systematic error. This was expected. Anthropometry is a man made collecting data so prone to systematic errors while bioimpedance is measured directly. The reliability of anthropometry calculation is supported by some similar studies on elderly community living individuals (Lee et al., 2001; Barbosa-Silva et al., 2016).

On the other hand, bioimpedance presented also high correlation to DXA but more specific than sensitive prone to diagnose predominantly negative (healthy) patients. In addition, Bland Altman's analysis on Bioimpedance showed higher bias (farther from the mean difference) and Student's T one test of bias was significant, different to the predicted mean difference (zero), supporting the lack of agreement to DXA for positive results (sick patient). On the other hand error dispersion was aleatory not suggesting proportional trend in errors. But the data accuracy measured by BIA is controversial in the literature. Beaudart *et al* (Beaudart et al., 2015) studied 250 outpatients, aged 74.1 ± 6.4, from a Belgium Clinical Health Service. They showed 8.4% to 27.6% prevalence of sarcopenia according to the BIA and DXA diagnostic criteria respectively. The lag between these percentages may reflect some limitations to the BIA method which were expected.

Equation by Jansen et al. for estimating skeletal muscle (SM) mass using bioelectrical impedance analysis (BIA) estimates the whole body SM mass determined by magnetic resonance imaging. Skeletal muscle mass was assessed using BIA, which has shown a good correlation with DXA and magnetic resonance imaging (Low et al., 2020). A wide range of techniques can be used to assess muscle mass. Cost, availability, and ease of use can determine whether the techniques are best suited to clinical practice or are more useful for research. No technique meets all requirements but DXA could be regarded as a reference standard for measuring muscle lean body mass (Buckinx et al., 2018).

Physiological alterations, such as poor hydration, higher inflammatory status resulting from comorbidities may affect the BIA analysis of resistance and reactance, DXA and anthropometric data are less affected by these variables (Lustgarten and Fielding, 2011; Doherty, 2001) In addition bioimpedance results depends on direct equipment measurements than on man's repeated collecting data so provides less systematic errors. BIA measurement reacts to instant shifting and changing of fluid status (Ritz and for the Source S, 2001). In our study, the hydration status of the elderly, was not strictly controlled, and the findings must have been affected by this variable. Only olders adults who presented compensated chronic diseases were included, but still, heart and kidney diseases can lead to biased data when measuring muscle mass and may have affected the BIA data in this study.

LoA should be judged mainly on clinical practice. This matter, clinical relevance was evident as the LoA in both anthropometry and BIA was adequate to predictive values from literature.

Future studies need to investigate the validity of these methods predictive of FFM, in other older populations, the chronically ill, people living in residential aged care and adults elderly of different ethnic groups and gender sex, before it can be generalized to all older people.

5. Limits of the study

This study includes only men, and, so, it is not possible to apply these data to the female population. But we opted for this study design to homogenize the population studied. In addition to that, this is a convenient population since it allows no external extrapolation. Another limitation is the lack of hydration assessment, since this factor impacts the all the body composition estimated by the methods used in this study. Another limiting issue is that age was not stratified in groups, since the population was not large enough to do so. We tried to reduce these limitations by inviting all the outpatients and not just a sample.

All measurements will have some degree of error and this must be taken into account when interpreting data and mostly when transferring the results to decision making process in clinical practice. We chose to use correlation tests, regression tests and Bland Altmn's analysis. The concept of high correlation must be set apart from that of causation. In many situations a strong relationship between variables may in fact be related to a third variable or several variables, which are associated to study the methods. This occurs more often in clinical observational studies, where some bias in data cannot be controlled.

There are some restrictions on the interpretation of linear regression analysis, which is based on the size of the residuals. We found a good agreement between methods. To minimize biased data, we chose to work with a specifically selected population. This maneuver limits the validation of the results, since predictions will not be applicable to those who do not meet the population criteria.

6. Conclusion

This vulnerable elderly miscegenate compensated outpatient with multiple diseases present clear signs of aging and strong data on sarcopenia. The assessment to LMM data was not more difficult because of aging or multiple diseases. Both BIA and anthropometry equations provided reliable measurements of LMM in these vulnerable patients. They proved to be accurate with high sensibility, specificity, predicted values and good agreement with DXA and may be applied independently. However, Anthropometry was highly sensitive and provided less biases (more agreement) with DXA. In addition, it also proved that it may substitute any other method of measuring LMM tested in this study, is worthwhile for vulnerable population, achievable by healthcare systems at lower cost, easily and quickly applicable in clinical practice in any place, regardless of the social and economic status, and trustworthy to support individual, collective, private and public health policies.

CRediT author statement

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Declaration of competing interest

The authors declare that there were no conflicts of interest.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.exger.2020.111020.

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